

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of M. Tech Mechanical (CAD/CAM) Theory Examination****November 2018****ME753 Advanced Machine Design****Date: 24/11/2018, Saturday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Use of scientific calculator is allowed.
4. Figures to right indicate the full marks.

SECTION – I**Q - 1 Do as directed.**

1. The process of converting a bulk load into a unit load is known as _____. **[01]**
2. Enlist two methods for improving lateral rigidity of the structure. **[02]**
3. Distinguish between fail-safe design and safe-life design. **[02]**
4. Positively corrected gear is known as S-Plus gear. True or False? **[01]**
5. When a part is subjected to a constant stress at high temperature for a long period of time, it will undergo a slow and permanent deformation called as _____. **[01]**
 a) Creep b) Resilience c) Fatigue d) None of these

Q – 2 (a) Discuss the basic objectives of material handling system. **[06]**

Q – 2 (b) A rectangular plate of size 150 mm × 100 mm with a central hole is subjected to a completely reversed axial load of 20 KN. The notch sensitivity can be assumed as 0.8. Determine the plate thickness for infinite life, if FOS is 2. Assume ultimate tensile strength as 500 MPa. The surface factor is 0.8, size factor is 0.85 and calculations are expected at 90% reliability, for which the reliability factor is 0.897. Take $K_t=2.5$ and $b/d=0.2$. **[08]**

OR

Q – 2 (b) A flat horizontal belt conveyor is used for transporting crushed rock having a mass density of 2 t/m³. The belt is 800 mm wide and has a speed of 1.75 m/s. Determine the capacity of the conveyor. The surcharge angle may be taken as 25°. The effective width b (in meters), of the material carried by the belt safely, is given by the following equation: $b = 0.9 B - 0.05$. **[08]**

Q – 3 (a) Define fracture. Explain different modes of fracture. [06]

OR

Q – 3 (a) Discuss on various phases for a typical design process. [06]

Q – 3 (b) Design a tensile bar of length $L = 200$ mm to carry a tensile load of 5 kN for minimum cost out of the following materials. [08]

Material	Mass Density (kg/m ³)	Material Cost (Rs/N Weight)	Yield Strength (MPa)
Steel	7500	16	130
AL-alloy	3000	32	50
Titanium-alloy	4800	480	90
Magnesium-alloy	2100	32	20

SECTION – II

Q - 4 Answer the following questions.

1. Compare concurrent engineering with traditional engineering. [04]
2. What is brainstorming? Explain it with two real-life examples. [03]
3. What is Design for manufacturability? [02]

Q – 5 (a) Discuss the factors for correcting endurance limit. [06]

OR

Q – 5 (a) Explain Johnson's method of optimum design. [06]

Q – 5 (b) A beam of size 100 mm × 25 mm × 10 mm having an edge crack in its central plane whose length is 5 mm. A load of 1000 N is applied opposite to crack so that crack would tend to open in bending. Calculate the SIF of crack. Find the SIF of crack when two loads of 500 N is placed at 25 mm from the central plane. [08]

Q – 6 Answer the following questions. (Attempt any two) [12]

1. State the different theories of failure. Explain any two in detail.
2. Differentiate between isotropic material and anisotropic material.
3. Prove, $x = \frac{Z_{\min} - Z}{Z_{\min}}$ for corrective gear design.
